

RESEARCH ARTICLE

A Generative Model for Cetacean Whistles Using Variational Autoencoder and Mixture of Gaussians

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This work was supported by the Future Challenge Defense Technology Research and Development Program of the Agency for Defense Development (ADD), funded by the Defense Acquisition Program Administration (DAPA), in 2026, under Grant 915084201.

ABSTRACT This paper proposes a tractable and compact generative model for cetacean whistle signals based on Variational Autoencoder (VAE) and mixture of Gaussians in underwater biomimetic communication. Cetacean whistles are nonlinear frequency-modulated signals with diverse time-frequency structures and can be interpreted as probability density functions. To learn the marginal probability distribution of whistle data, this study incorporates a VAE and a mixture of Gaussians. The VAE learns latent space representations from whistle spectrogram images, and the prior distribution of the latent variables is modeled using multiple Gaussian components. Since the latent space of whistle contour is modeled as an explicit density function of multimodal structure, it enables unsupervised clustering and conditional whistle generation by analyzing component-wise features. The statistical distance between the synthesized whistles and the training data was evaluated using the Fréchet inception distance, Recall, Precision, and F1-score yielding a score of 2.6767, 87.21%, 91.99%, and 89.54%, respectively, when the number of Gaussian components was set to 10,000.

INDEX TERMS Cetacean whistle, conditional generative model, generative model, mixture of Gaussians, variational autoencoder, biomimetic underwater communication.

I. INTRODUCTION

A. BRIEF INTRODUCTION OF BIOMIMETIC UNDERWATER COMMUNICATION

Underwater biomimetic communication is a technique that emulates acoustic signals produced by marine mammals, primarily utilizing cetacean clicks and whistles [1], [2], [3], [4], [5], [6], [7], [8], [9]. Cetacean whistles are nonlinear frequency-modulated signals characterized by time-varying frequencies, which appear as diverse contours in spectrograms. Previous studies have proposed communication schemes that replicate the structural patterns of whistle contours or have adapted conventional digital modulation techniques for biomimetic purposes [1], [2], [3], [4], [5], [6], [7]. In contrast, clicks are broadband pulse

signals, and to fully exploit their wideband characteristics, time-delay modulation—where information bits are encoded in the time intervals between pulses—has been predominantly used [8], [9].

However, click sounds are generally unsuitable for biomimetic communication due to their broadband characteristics [37]. This limitation arises for two primary reasons. First, click signals have a relatively short propagation range. In particular, their high-frequency components experience significant attenuation due to underwater transmission loss, making them unsuitable for long-range communication over several kilometers. In contrast, whistles—with their relatively narrowband nature—are more favorable for long-distance transmission. Second, there are design constraints in underwater acoustic projectors. Most current projectors have difficulty generating very short-duration broadband pulses effectively.

The associate editor coordinating the review of this manuscript and approving it for publication was Sotirios Goudos³.

Biomimetic communication enables underwater platforms to exchange information covertly. Communication signals that mimic the characteristics and structure of biological sounds may be detected or intercepted by non-cooperative platforms; however, they are misidentified as biological emissions and subsequently ignored. Therefore, the precise modeling of the probability density function of biological signals is essential to ensure the covert nature of biomimetic communication. In this context, it is important to first collect a sufficient number of biological signal samples x , estimate their marginal probability distribution $p(x)$, and then generate synthetic communication signals that follow the estimated distribution.

B. MOTIVATION OF THIS WORK

Whistle-based biomimetic communication requires the acquisition of a large dataset of whistle contours from the target species to preserve covertness. The communication signal can then be constructed by either selecting a finite set of representative whistles for modulation or by employing generative methods that replicate the underlying probability distribution of real whistles. Nevertheless, some biomimetic communication approaches are inherently limited in their ability to faithfully mimic real whistle contours. For example, the modulation scheme called Binary Orthogonal Keying (BOK), proposed in [10], divides a whistle signal into fixed symbol intervals and modulates each symbol using two types of chirp signals, each centered around the whistle frequency of the corresponding interval. However, such approaches that modify the original whistle signal may reduce its similarity to real whistle signals, posing a limitation in terms of biomimicry.

In contrast, there have been efforts to utilize real whistle contours directly. Ahn et al. [11] extracted and selected whistle signals from white-sided dolphins and applied them to a machine learning-based biomimetic communication framework. In their study, the whistles were grouped, and communication symbols were assigned based on the transitions between these groups. A total of 704 selected whistles were used in the system.

However, repeating a limited set of whistle patterns using underwater acoustic projectors can compromise the covert nature of biomimetic communication. Deterministic methods that select and transmit from a fixed set of signals may allow adversarial systems—designed to continuously monitor and analyze biological acoustic emissions—to identify recurring patterns. Such deterministic signal generation approaches increase the risk of exposing not only the presence of biomimetic communication but also its modulation schemes and system parameters. The advancement of deep learning-based artificial intelligence technologies further amplifies this risk.

One approach to mitigating the limitations of using fixed whistle patterns, as mentioned earlier, is to model the whistle shapes as a probability density function and to utilize newly generated data sampled from this distribution. Although the

sampled whistles resemble the original patterns, they are not exact replicas, making them more likely to be perceived as natural biological signals.

Accordingly, the objective of this study is to design a deep generative model for biomimetic underwater communication that can be deployed on surface or underwater platforms and that faithfully replicates the whistle sounds of observed whales or dolphins. Specifically, the model is designed to follow the estimated probability distribution of the collected whistle data. By leveraging the proposed approach, there is a higher chance that non-cooperative platforms will interpret the generated signals as genuine biological emissions, thereby decreasing the risk of the transmitting platform being exposed.

C. RELATED WORK

Considering whistle data in the image domain, such as spectrograms, the whistle generation technique addressed in this study can be implemented using widely adopted deep generative models [12], [13], [14], [15], [16]. Currently, commonly used models for image generation include energy-based models [12], variational autoencoders (VAEs) [13], normalizing flows [14], generative adversarial networks (GANs) [15], and diffusion models [16], [17].

To the best of our knowledge, GANs are the only approach of deep learning-based generative model that has been applied to cetacean whistle generation [18], [19], [20]. These GAN-based methods were introduced to avoid repeated use of identical whistle sounds or to mitigate the issue of limited training data [18], [19], [20]. Zhang et al. [18] trained a Deep WaveGAN to generate waveform-level bottlenose dolphin whistles and evaluated the similarity between synthetic and real data using various time- and frequency-domain metrics. Wang et al. [19] addressed the data scarcity problem by applying various augmentation techniques to a small set of bottlenose dolphin whistles and trained a masking autoencoder and GAN to generate high-quality signals. Kim et al. [20] proposed WhistleGAN, which combines Mel filter banks with a deep convolutional GAN, to avoid the repetition of whistle sounds in biomimetic communication and presented a computer simulation for biomimetic communication based on synthetic biological signals. All three studies [18], [19], [20], which rely on the inherent characteristics of GANs, enable the generation of high-quality whistle data at relatively high speed.

However, GANs suffer from two fundamental limitations. The first is mode collapse. Because GANs generate data through a minimax game between the generator and the discriminator, the generator may learn to produce only a narrow subset of data that can fool the discriminator [21]. Mode collapse is a well-known issue intrinsic to GANs, and thus incorporating them into the transmission architecture of biomimetic communication introduces practical constraints. Second, GANs do not explicitly model probability distributions. As a result, even after training a whistle generation model, the underlying distribution remains intractable.

In contrast, explicit modeling of probability distributions enables additional capabilities such as density estimation, likelihood calculation, and unsupervised clustering—features that GANs do not inherently support.

The VAE and Mixture of Gaussians-based approach proposed in this study has not been utilized in the bio-signal generation field but has primarily been considered within the machine learning community [22], [23], [24], [25], [26], [27], [28], [38], [39]. Specifically, modeling the prior distribution $p(z)$ of the latent variables in a VAE using a multimodal structure such as a mixture of Gaussians enhances the expressiveness of the generative model for complex data, enables unsupervised clustering. Dilokthanakul et al. [22] proposed the Gaussian Mixture Variational Autoencoder (GMVAE), which models the latent prior as a Gaussian Mixture Model (GMM) and introduces both discrete and continuous higher-level latent variables, enabling the model to capture natural and interpretable cluster structures in an unsupervised setting. They also analyzed the issue of cluster degeneracy caused by over-regularization in standard VAEs and introduced a simple heuristic to effectively mitigate this problem. Jiang et al. [23] proposed Variational Deep Embedding (VaDE), an unsupervised generative clustering framework that incorporates a GMM prior into VAE architecture, enabling joint learning of latent representations and cluster assignments while retaining the model's generative capacity.

However, the approaches presented in [22], [23], [24], [25], [26], [27], and [28] require a rapidly increasing number of learnable parameters as the number of clusters grows. For the large-scale clustering level (approximately 10,000 clusters) considered in this study, such methods become computationally inefficient. For example, both GMVAE and VaDE simultaneously learn a mean vector and a covariance matrix for each cluster during network training, resulting in a substantial increase in learnable parameters as the number of clusters rises. Similarly, VampPrior must learn a pseudo-input corresponding to each cluster, with the same dimensionality as the input image, which further increases the parameter count. Although these existing methods may effectively capture the latent space when using a relatively small number of clusters, employing a larger number of clusters is more advantageous for achieving fine-grained and flexible conditional generation.

D. CONTRIBUTIONS OF THE STUDY

This paper proposes a VAE-GMM framework for modeling the marginal probability distribution of whistle image data from false killer whales (*Pseudorca crassidens*). The whistle data generated by the proposed VAE-GMM exhibit a high degree of similarity to real biological signals. When evaluated using the Fréchet Inception Distance (FID), Recall, and Precision metrics, the proposed model outperformed existing methods while maintaining a comparable generation speed. Unlike traditional biomimetic communication approaches

that repeatedly transmit a limited set of playback signals, the proposed VAE-GMM based whistle generation method enables a more authentic biomimetic reproduction of real biological sounds.

Furthermore, because the VAE-GMM organizes data according to the morphological characteristics of whistles, it inherently provides the effect of unsupervised clustering. During training, the VAE maps whistles with similar structural patterns to neighboring regions in the latent space. When this latent space is subsequently modeled using a GMM, similar patterns are grouped into the same Gaussian component.

Moreover, the proposed VAE-GMM supports highly flexible conditional whistle generation. Since each Gaussian component—or a group of components—can represent a specific whistle shape, generating desired whistle types becomes feasible by sampling from specific components. Conditional generation becomes feasible by analyzing the whistle characteristics such as whistle length, minimum and maximum frequency, start and end frequency, bandwidth, and center frequency. By analyzing other whistle characteristics in advance, the model can be easily extended to generate new samples under specific conditions.

The conditional generation mechanism of the VAE-GMM offers two key advantages over existing approaches such as Conditional GANs (CGANs) and Conditional Diffusion Models (CDMs) that employ classifier guidance or classifier-free guidance [40], [41]. First, VAE-GMM does not require explicit class label assignments for conditional generation. By modeling the latent space with a GMM, the images are automatically clustered and categorized. In contrast, CGANs and CDMs must be trained with predefined class labels as conditioning inputs. Since whistle patterns exhibit highly diverse and continuous variations, it is practically infeasible to predefine discrete classes for them, which further highlights the advantage of VAE-GMM in eliminating the need for class assignment. Second, VAE-GMM demonstrates superior flexibility in conditional generation. In CGANs and CDMs, the predefined labels serve as the sole mechanism for controlling conditional synthesis, which inherently limits their ability to generalize to diverse generation conditions. In contrast, the proposed method can flexibly generate data under various conditions by analyzing the characteristics of each Gaussian component.

The proposed method is clearly distinguished from existing cetacean whistle generation models [18], [19], [20] in that it enables flexible and scalable conditional generation. Since the models in [18], [19], and [20] are based on GANs, their conditional generation mechanism is limited to CGANs, and their underlying probability distributions remain intractable.

In contrast, the relatively simple combination of VAE and GMM offers greater flexibility and efficiency in conditional generation, primarily due to the characteristics of the whistle image dataset. Unlike large-scale vision datasets such as ImageNet or COCO, the whistle image dataset contains less information. Although the whistle contours are diverse in



FIGURE 1. Examples of whistle image data used to train the proposed VAE-GMM model: The actual data used in this study are normalized to have intensities between 0 and 1, with the whistle regions having values close to 1. For better visual clarity, the intensities have been inverted in the figure.

shape, the images are grayscale with almost no background content, making lightweight models more suitable for this domain. Therefore, we argue that the proposed VAE-GMM approach is more appropriate than current state-of-the-art diffusion models, particularly in terms of conditional generation flexibility and the tractability of the probability distribution.

The key contributions of this study are summarized as follows:

- 1) The VAE-GMM is shown to effectively model the marginal distribution of whistle images, and its generation performance is evaluated using the FID, Recall, and Precision.
- 2) We model the latent variables of whistle images using the GMM and perform a component-wise analysis. By doing so, we demonstrate that flexible and extensible conditional whistle generation is achievable for various parameters—such as signal length, maximum/minimum frequency, and bandwidth—without the need for explicit class labels.
- 3) The VAE-GMM is shown to support downstream tasks such as unsupervised clustering through MAP estimation.
- 4) We proposed a lightweight architecture specifically optimized for whistle generation to achieve high computational efficiency and throughput, thereby demonstrating its technical feasibility for resource-constrained real-time applications.

E. OUTLINE

The remainder of this paper is organized as follows. Section II describes the dataset and the training procedure of the VAE-GMM. Section III presents the methods for marginal and conditional sampling, and unsupervised clustering. Section IV describes experimental results and a comprehensive performance analysis. Section V offers a detailed discussion of the study's limitations. Finally, Section VI concludes the paper.

II. LEARNING WHISTLE GENERATIVE MODEL

A. PROBLEM FORMULATION

In this work, we consider the problem of modeling dolphin whistle signals for the purpose of biomimetic sound source

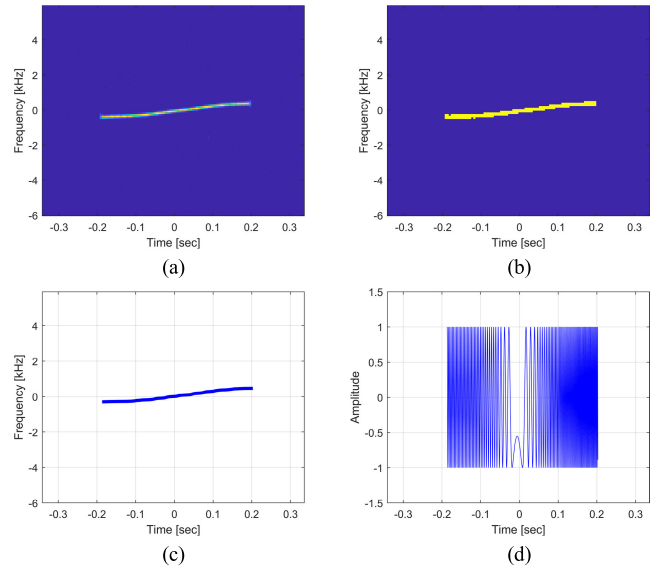


FIGURE 2. Process of generating an NFM signal from a whistle image: (a) the first panel shows an example of a whistle image generated by the VAE-GMM, (b) by applying a simple threshold of 0.05 to the image, the whistle contour shown in the second panel can be obtained, (c) the third panel presents the estimated time-frequency vector derived from the binary whistle image consisting of 0s and 1s, (d) the final panel displays the resulting constant-amplitude NFM signal generated based on this time-frequency vector.

generation and underwater covert communication. In general, cetacean whistles are observed as continuous

sequences over time. Let the observed whistle sequence be denoted as a vector $\mathbf{x} = (x_1, x_2, \dots, x_N)$, where each x_i represents an individual whistle signal and N is the number of whistles.

Assuming that the whistles are generated independently and identically from an unknown probability distribution $p(x)$, the joint probability of the whistle sequence $\mathbf{x}$ can be factorized as:

$$p(\mathbf{x}) = \prod_{i=1}^N p(x_i) = p(x)^N. \quad (1)$$

With the i.i.d. assumption, modeling an entire whistle sequence becomes a task of learning the distribution of a single whistle, $p(x)$. Therefore, the main objective of this study is to construct a biomimetic generative model that accurately captures the underlying distribution $p(x)$ of individual whistle signals. In this sense, modeling the probability distribution of an individual whistle is equivalent to modeling the sequence-structured whistle data we have observed.

The distribution $p(x)$ can be represented by introducing a latent variable z , which signifies an underlying factor capable of explaining the whistle data as follows:

$$p(x) = \int p(x, z) dz = \int p(x|z) p(z) dz. \quad (2)$$

However, since $p(x)$ is practically intractable, the whistle generative model is implemented by learning an approximate posterior distribution $q(x|z)$ of $p(x|z)$ and modeling prior distribution of latent vector $p(z)$ with a mixture of Gaussians.

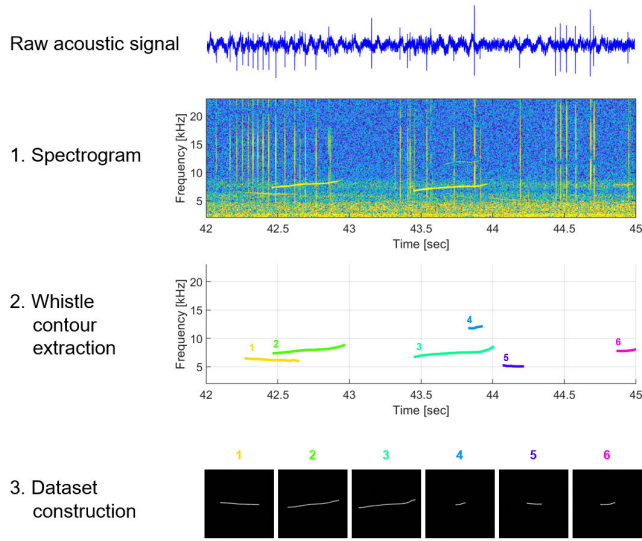


FIGURE 3. Preparation process of the whistle dataset: The raw acoustic signal is first converted into a spectrogram, and a whistle contour extraction technique is then applied to obtain the time-frequency vector of each individual whistle. The time-frequency vector is subsequently transformed into a whistle image with the center frequency removed through constant-amplitude NLFM and spectrogram conversion processes.

This formulation enables an approach to generating realistic dolphin-like whistles and provides a foundation for applications such as biomimetic underwater communication and the development of data-driven signal synthesis techniques.

B. DATA REPRESENTATION

The domain of the target whistle data x can take various forms. In this study, we adopt a representation in which each whistle is expressed as a single image depicting the contour of a baseband whistle signal with the center frequency removed as in Fig. 1.

The time-frequency contour information embedded in the image-based whistle data can be easily converted into a time-frequency vector using simple thresholding techniques. Based on this vector, the final nonlinear frequency-modulated (NFM) signal can be synthesized. Fig. 2 illustrates the process of extracting a time-frequency vector from a single-whistle image using thresholding, and generating a constant-amplitude NFM signal based on this vector.

C. WHISTLE DATASET

To construct a whistle spectrogram dataset for the false killer whale, this study utilized the Detection, Classification, Localization and Density Estimation (DCLDE) 2022 Raw Passive Acoustic Data, which was publicly released online by the National Oceanic and Atmospheric Administration (NOAA) [29]. The dataset comprises underwater acoustic recordings collected using a multi-channel towed hydrophone array during the Hawaiian Islands Cetacean and Ecosystem Assessment Survey (HICEAS) conducted between July and November 2017. The recordings were obtained alongside

visual line-transect surveys aimed at estimating cetacean abundance and density.

For this study, we selected 2,486 acoustic files containing whistles from the false killer whale as the target data. Each file generally contains 60 seconds of underwater acoustic signals measured from six hydrophones arranged in a towed array configuration, and the sampling rate is 500 kHz.

Fig. 3 illustrates the overall process of constructing the whistle dataset from raw acoustic signals. This process consists of three main stages: spectrogram conversion, whistle extraction, and dataset preparation.

In the first stage, the raw acoustic signals containing whistles are converted into spectrograms, allowing the time-frequency characteristics of the whistles to be visually identified. The spectrogram was generated after resampling the signal to 192 kHz, applying a 2,048-point Hamming window, and using a 50% overlap. As a result, the time and frequency resolutions were approximately 5.3 ms and 93.75 Hz, respectively. To suppress low-frequency ocean noise, a high-pass filter with a passband above 2 kHz was applied.

In the second stage, a neural network-based whistle extraction method [31] was applied to detect individual whistle contours. The extraction process was combined with a post-processing algorithm [32], [33] that converts detection outputs into individual whistle tracks.

In the final stage, the extracted whistle time-frequency vectors were converted into individual whistle images to construct the final training dataset. A total of 228,673 whistle time-frequency vectors were obtained from 2,486 raw acoustic signal files. Each vector was then used to generate a constant-amplitude NFM baseband signal and its spectrogram, which was further transformed into a 128×128 grayscale image (See Figs. 1 and 3).

D. LEARNING VARIATIONAL AUTOENCODER

VAEs are probabilistic generative models that learn a latent variable representation of data. The VAE consists of an encoder network that approximates the intractable true posterior distribution $p(z|x)$ with a tractable variational distribution $q(z|x)$, and a decoder that models the conditional likelihood $p(x|z)$. The model is trained by maximizing the Evidence Lower Bound (ELBO), which jointly encourages accurate reconstruction of the data and regularization of the latent space via the KL divergence between the approximate posterior and the prior $p(z)$.

To control the strength of latent regularization, a weighting factor β can be applied to the regularization term $\mathcal{L}_{KL}$ in the VAE loss. The total loss function $\mathcal{L}(x)$ is then given by the following expression:

$$\mathcal{L}(x) = \mathcal{L}_{rec} + \beta \mathcal{L}_{KL}, \quad (3)$$

where $\mathcal{L}_{rec} = -\mathbb{E}_{q(z|x)} [\log p(x|z)]$ and $\mathcal{L}_{KL} = KL(q(z|x) \parallel p(z))$. Here, $\mathbb{E}(\cdot)$ denotes the ensemble average and $\mathcal{L}_{rec}$ denotes the reconstruction loss.

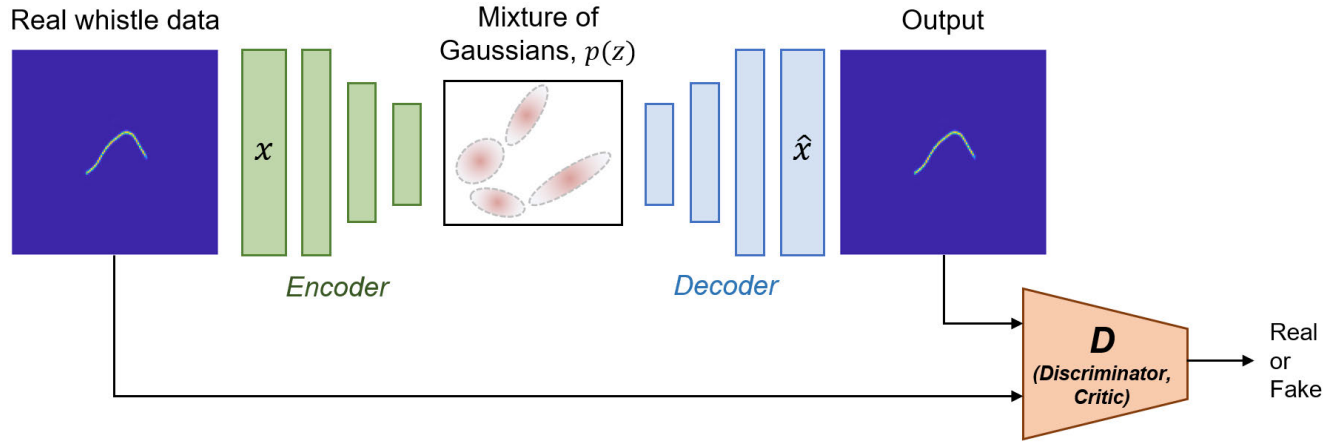


FIGURE 4. VAE learning framework integrated with a WGAN-GP critic. The encoder functions as a variational posterior that maps whistle images into latent variables, which are modeled by a GMM. The decoder reconstructs the whistle images from these latent variables, while the WGAN-GP critic ensures that the output of the VAE aligns with the actual data distribution.

Fig. 4 illustrates the training framework of the VAE. The input data x is encoded into a latent variable z through the encoder, and the decoder is trained to reconstruct the input by generating an output $\hat{x}$ that closely resembles the original input. In (3), by setting the weighting coefficient $\beta \ll 1$, the constraint imposed by the prior distribution of latent vector $z \sim \mathcal{N}(0, I)$ can be relaxed to some extent, allowing the latent space to be represented on a specific manifold rather than as a general multivariate normal distribution. $z \sim \mathcal{N}(0, I)$ can be relaxed to some extent, allowing the latent space to be represented on a specific manifold rather than as a general multivariate normal distribution.

As shown in Fig. 4, we trained an additional critic network to encourage the VAE outputs to resemble real images more closely. This training framework is a modified version of the autoencoder training scheme used in latent diffusion models [17], where we added an adversarial loss term in addition to the reconstruction loss and the KL divergence in (3). To ensure stable training, both the VAE and the critic were optimized using the Wasserstein GAN with Gradient Penalty (WGAN-GP) [49]. A weighting coefficient λ was introduced to balance the adversarial loss $\mathcal{L}_{adv}$, and the final loss function $\mathcal{L}_{\beta-VAE}(x)$ used for VAE training in this study is defined as follows:

$$\mathcal{L}_{\beta-VAE}(x) = \mathcal{L}_{rec} + \beta \mathcal{L}_{KL} + \lambda \mathcal{L}_{adv}, \quad (4)$$

where $\mathcal{L}_{adv} = -\mathbb{E}_{z \sim q(z|x)} [\log D(p(x|z))]$ and $D(\cdot)$ denotes the output of the critic network. The critic follows the standard WGAN-GP loss function $\mathcal{L}_D(x, \hat{x})$, defined as:

$$\mathcal{L}_D(x, \hat{x}) = \mathbb{E}_{\hat{x} \sim P_{VAE}} [D(\hat{x})] - \mathbb{E}_{x \sim P_r} [D(x)] + \lambda_{GP} \mathbb{E}_{\tilde{x}} \left[(\|\nabla_{\tilde{x}} D(\tilde{x})\|_2 - 1)^2 \right], \quad (5)$$

where P_{VAE} and P_r denote the distributions of the VAE outputs and the real data, respectively. $\hat{x}$ is the VAE outputs, $\tilde{x}$ is defined as $\tilde{x} = \epsilon x + (1 - \epsilon) \hat{x}$ for $\epsilon \in U(0, 1)$, and λ_{GP} is the weighting coefficient for the gradient penalty.

The adversarial loss in (4) differs from the conventional loss in that it applies a logarithmic function to the critic's output. We observed that applying the log function to $\mathcal{L}_{adv}$ led to improved performance during the early stages of VAE training. This effect is considered to be similar to alleviating the vanishing gradient problem, much like replacing the generator loss in GANs from $\min(1 - \log D)$ to $\max(\log D)$ [34]. The network architectures and the training configuration of the VAE encoder, decoder, and critic are described in the Appendix A.

E. LEARNING GAUSSIAN MIXTURE MODEL

The objective of training the GMM model for whistle generation is to effectively model the latent manifold using K Gaussian components. In other words, the goal is to estimate the parameters of these Gaussian distributions. Accordingly, the prior distribution of the latent variable can be expressed as a linear combination of K Gaussian components:

$$p(z) = \sum_{k=1}^K \pi_k \mathcal{N}(z | \mu_k, \Sigma_k), \quad (6)$$

where π_k denotes the mixing coefficient (a parameter of the categorical distribution), μ_k is the mean vector, Σ_k is the covariance matrix, and $\mathcal{N}(z | \mu_k, \Sigma_k)$ represents the k -th Gaussian component. The parameter π_k must satisfy $0 \leq \pi_k \leq 1$, and $\sum_k \pi_k = 1$.

Suppose we have a set of latent representations $\{z_1, z_2, \dots, z_N\}$ extracted from the whistle dataset. Based on these samples, the goal is to estimate the parameters π , μ , and Σ of the mixture of Gaussians. These parameters can be effectively estimated using the well-known Expectation-Maximization (EM) algorithm [35]. In this study, we adopt the standard EM algorithm to learn the parameters of the prior distribution $p(z)$ (See details in Appendix B). The input for GMM training consists of the latent mean outputs of encoder.

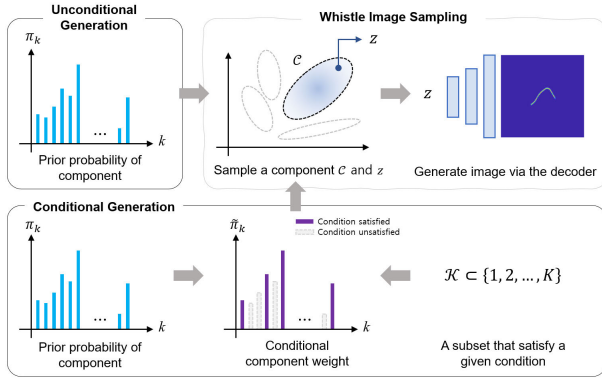


FIGURE 5. Process of unconditional and conditional whistle generation: In unconditional generation, the prior probability of each Gaussian component is directly used for data sampling. In conditional generation, the component weights are converted and normalized into conditional component weights according to the given condition before being applied to sampling. After sampling a component and a latent vector, the latent vector is fed into the decoder to synthesize the corresponding whistle image.

III. SAMPLING AND INFERENCE METHODS

This section describes the methods for unconditional and conditional whistle generation, and unsupervised clustering based on MAP estimation.

A. UNCONDITIONAL GENERATION

After sequentially training the VAE and GMM, unconditional whistle generation can be sequentially performed by sampling $\mathcal{C}$, z , and $\hat{x}$ as shown in Fig. 5. The unconditional generation process consists of the following three steps:

- 1) Sample a Gaussian component $\mathcal{C} \sim \text{Categorical}(\pi)$.
- 2) Sample $z \sim \mathcal{N}(z|\mu_{\mathcal{C}}, \Sigma_{\mathcal{C}})$.
- 3) Generate $\hat{x} \sim p_{\theta}(x|z)$, where $p_{\theta}(x|z)$ represents the VAE decoder network.

B. CONDITIONAL GENERATION

Conditional generation refers to the process of synthesizing whistle signals that satisfy specific conditions, where each Gaussian component—or a group of components—may represent a particular whistle shape. To generate target shape effectively, a conditional generation method that is both controllable and tractable is essential. The VAE-GMM framework proposed in this study is well-suited to this task, as it explicitly models the prior distribution of latent variables.

Conditional generation of whistle sounds can be interpreted as the process of sampling a latent variable z from a subset $\mathcal{K} \subset \{1, 2, \dots, K\}$ of components that satisfy a given condition. Accordingly, the latent variable z can be sampled from:

$$z \sim \sum_{k \in \mathcal{K}} \tilde{\pi}_k \mathcal{N}(z|\mu_k, \Sigma_k), \quad (7)$$

where

$$\tilde{\pi}_k = \frac{\pi_k}{\sum_{k \in \mathcal{K}} \pi_k}. \quad (8)$$

Equation (8) represents a normalized form of the mixing coefficients of the Gaussian components that satisfy the

condition. In this paper, $\tilde{\pi}_k$ is defined as the conditional component weight.

Once the relevant components are identified, this normalization can be easily computed. Based on (7) and (8), the conditional generation process as shown in Fig. 5, similar to unconditional generation, consists of the following three steps:

- 1) Sample a Gaussian component $\mathcal{C} \sim \text{Cat}(\tilde{\pi})$ in $\mathcal{K}$.
- 2) Sample $z \sim \mathcal{N}(z|\mu_{\mathcal{C}}, \Sigma_{\mathcal{C}})$.
- 3) Generate $\hat{x} \sim p_{\theta}(x|z)$.

This conditional generation method assumes that the selected Gaussian components are suitable for generating whistle sounds that meet the specified condition. In practice, multiple whistle latent samples $z \sim \mathcal{N}(\mu_{\mathcal{C}}, \Sigma_{\mathcal{C}})$ can be generated from each component $\mathcal{C}$ and analyzed to determine whether the component satisfies the condition. For example, let $l(x)$ denote the duration of a whistle sound. Prior analysis is required to verify whether the whistles generated from a specific component satisfy $L_1 \leq l(x) \leq L_2$, where L_1 and L_2 are minimum and maximum length condition, respectively. If the condition is satisfied, the component can be included in the subset $\mathcal{K}$.

Therefore, once the VAE and GMM have been trained, it is necessary to analyze the distribution of feature vectors of the whistle samples $x \sim p(x|\mathcal{C})$ generated from each component. The feature vector can be designed to capture various properties of whistle sounds. In this paper, we consider the minimum and maximum frequencies, start and end frequencies, bandwidth, whistle length, and center frequency. We selected seven commonly used statistical features in the field of bioacoustics for analyzing cetacean whistle signals [42], [43], and the types of these features can be easily extended if necessary.

However, even for components that satisfy the target condition, some generated samples may fail to meet the requirement precisely. For instance, when aiming to generate whistle sounds with durations between 30 and 40 samples, samples with lengths ranging from 28 to 42 may still appear in a specific component.

In such cases, whistle samples that do not satisfy the condition can be discarded to form the desired conditional probability distribution. This strategy can be interpreted as a form of rejection sampling [35].

Furthermore, by analyzing the acceptance ratio of samples generated from each component, the mixing coefficients of the overall conditional probability distribution can be approximated. The acceptance ratio a_k ($0 \leq a_k \leq 1$) is defined as the proportion of whistle samples generated from $\mathcal{C}_k$ that satisfy the given condition. Then the conditional latent distribution $p(z|\mathcal{K})$ and adjusted conditional component weight $\tilde{\pi}_k$ can be calculated as follows:

$$p(z|\mathcal{K}) = \sum_{k \in \mathcal{K}} \tilde{\pi}_k \mathcal{N}(z|\mu_k, \Sigma_k), \quad (9)$$

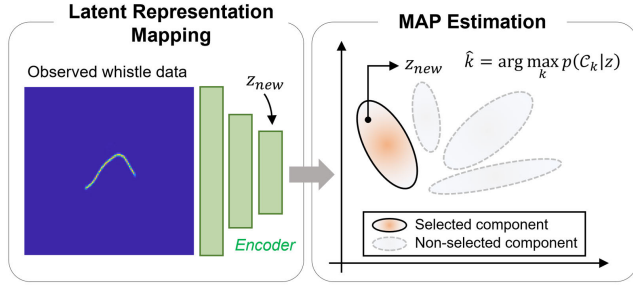


FIGURE 6. Unsupervised clustering process: new whistle data are fed into the trained VAE encoder to obtain latent vectors, and the MAP estimation assigns the input data point to its most probable Gaussian component.

where

$$\tilde{\pi}_k = \frac{a_k \pi_k}{\sum_{k \in \mathcal{K}} a_k \pi_k}. \quad (10)$$

C. UNSUPERVISED CLUSTERING

As shown in (6), the proposed model explicitly defines the marginal distribution of the latent variables. Given a new whistle observation x_{new} and its corresponding latent vector z_{new} obtained from the VAE encoder of the approximated posterior $q(z|x)$, it becomes possible to compute the responsibility $p(\mathcal{C}|z)$ for each Gaussian component. The responsibility represents the posterior probability that z_{new} belongs to a particular Gaussian component, thereby indicating which component best matches the observed whistle data. This can be computed using Bayes' theorem as follows:

$$p(\mathcal{C}|z) = \frac{p(\mathcal{C})p(z|\mathcal{C})}{p(z)} = \frac{\pi \mathcal{N}(z|\mu, \Sigma)}{\sum \pi \mathcal{N}(z|\mu, \Sigma)}, \quad (11)$$

where π is $p(\mathcal{C})$ and $p(z|\mathcal{C})$ denotes the conditional probability density of multivariate Gaussian distribution.

Fig. 6 illustrates a conceptual example in which the responsibilities for a new input x_{new} are computed. The responsibilities are evaluated for all Gaussian components, and the component with the highest posterior probability is selected as the most likely generator of the input. This MAP estimation enables unsupervised clustering and is expressed as:

$$\hat{k} = \arg \max_k p(\mathcal{C}_k | z). \quad (12)$$

This approach allows effective clustering without requiring a separate discriminative model. In fact, the proposed VAE-GMM represents a generative model applied to a discriminative task, which has also been discussed in [37, Ch. 1, pp. 43]. We further anticipate that MAP estimation can be

used to estimate which Gaussian component was likely responsible for generating a received whistle signal at the receiver side.

IV. ANALYSIS OF WHISTLE GENERATIVE MODEL

This section presents the results of unconditional and conditional whistle generation using the proposed VAE-GMM-based generator, as well as corresponding performance analyses. The evaluation focuses on both the

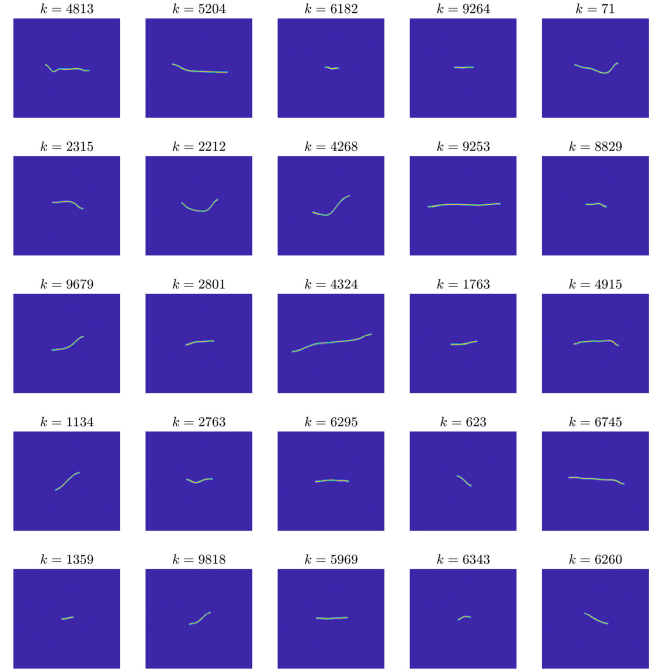


FIGURE 7. Unconditional whistle image synthesis results and the corresponding Gaussian component index (VAE-64, $K = 10,000$).

reconstruction output of the VAE from the training dataset and the generated whistle images from the VAE-GMM model. In addition, we provide examples of unsupervised clustering based on the MAP estimation.

A. UNCONDITIONAL WHISTLE GENERATION EXAMPLES

Unconditional whistle generation follows the procedure described in Section III-A. Fig. 7 visualizes the results of sampling whistle images from various Gaussian components of the VAE-64-based VAE-GMM model with $K = 10,000$ and $\mathcal{L} = 64$, together with their corresponding component indices, where K and $\mathcal{L}$ represent the number of Gaussian components and the latent vector dimension, respectively. As illustrated, the proposed model can effectively generate diverse and high-quality whistle images.

B. CONDITIONAL WHISTLE GENERATION PROCESS

1) COMPONENT-WISE FEATURE ANALYSIS

Before performing conditional whistle generation, it is essential to analyze the characteristics of whistle feature generated by each component of the mixture of Gaussians. To this end, a large number of whistle samples are generated from each component, and the statistical distributions of their extracted feature vectors are analyzed to identify the generative behavior of each component.

Fig. 8 presents whistle samples repeatedly generated from four components (indices $k = 7,124, 4,268, 1,253$, and $6,114$) in the VAE-GMM model with $K = 10,000$ and whistles tend to terminate at their peak frequency. Such comparisons between feature distributions can be quantitatively assessed using the KL divergence, which is left $\mathcal{L} = 64$, along with a detailed analysis of the corresponding

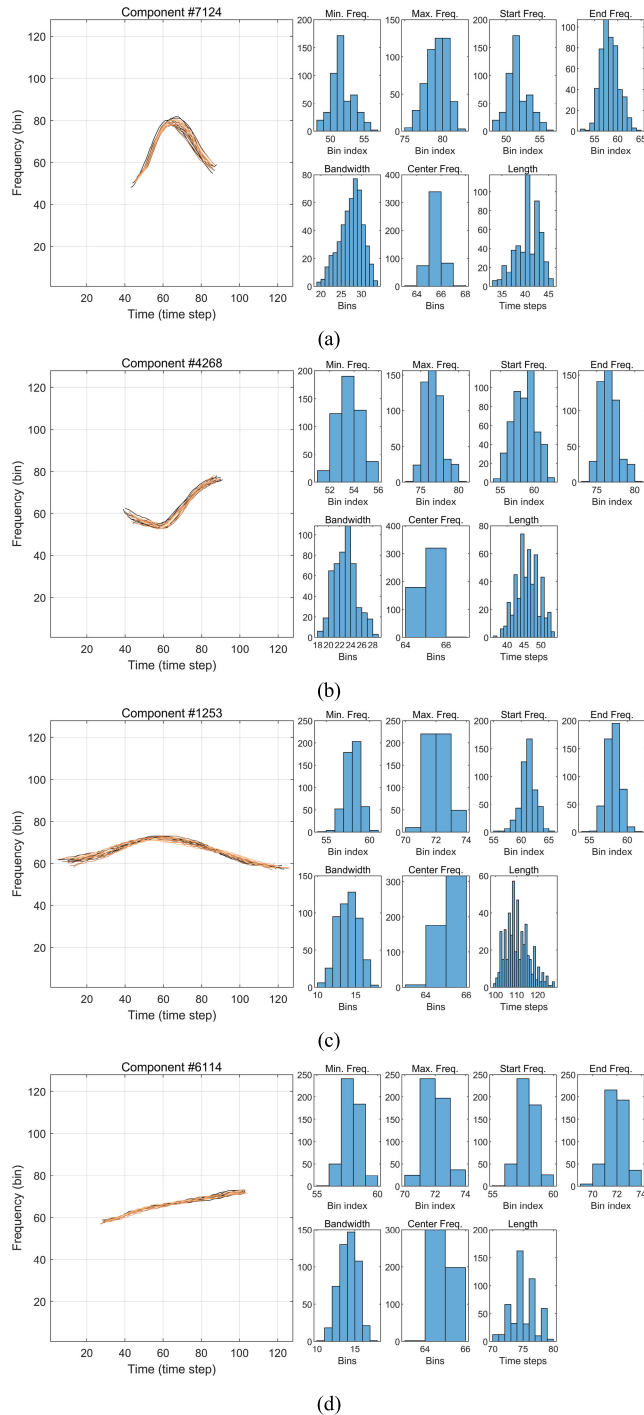


FIGURE 8. Whistle generation results generated by each Gaussian component and corresponding feature vector distributions. The left panel shows an overlaid visualization of 20 whistle contours generated from the same component, while the right panel presents histograms of seven feature vector distributions derived from 500 generated whistles for each component. The results were generated using a VAE-64 model and a GMM with $K = 10,000$, with component indices as follows: (a) $k = 7, 124$, (b) $k = 4, 268$, (c) $k = 1, 253$, (d) $k = 6, 114$.

feature vectors. As shown, whistle signals generated from the same component exhibit similar structural characteristics, and their feature vectors tend to follow consistent statistical distributions. For example, the average lengths of whistles generated from components 7,124 and 1,253 are

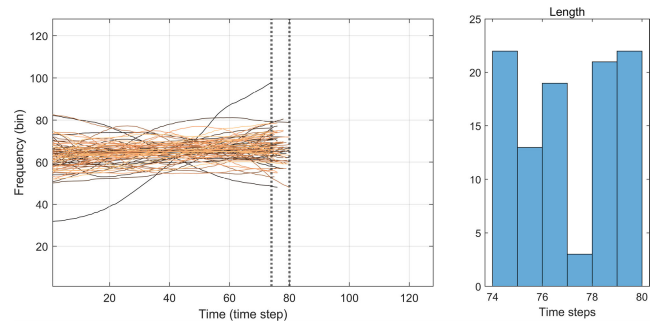


FIGURE 9. Conditional whistle generation examples. The left panels show the overlaid visualization of 100 whistle samples that satisfy each specified condition, with the dotted lines indicating the boundaries of the condition range. The right panels present histograms of feature vectors showing the distribution of the generated whistles according to the specified length condition of 74-80 samples.

approximately 40 and 110 samples, respectively, indicating a clear distinction between them.

Moreover, analyzing differences between the distributions of different features within the same component can provide additional insights into the internal characteristics and generation tendencies of each component. For instance, in Fig. 8(b), the distributions of maximum frequency and end frequency nearly the identical, suggesting that the generated for future work.

Through this component-wise statistical analysis, one can construct a subset of components that satisfy specific conditions and perform conditional whistle generation accordingly.

2) CONDITIONAL GENERATION RESULTS

Fig. 9 illustrate the results of conditional whistle generation based on constraints on duration. In the left panel overlays 100 whistle samples that meet the specified condition, with the dashed lines indicating the boundary of the target condition range. As shown, a substantial number of generated samples conform to the given conditions. The right panel displays histograms of the corresponding features, which also demonstrate that the generated whistles satisfy the specified criteria. This approach can be further extended to conditional generation based on other whistle characteristics as needed. Additional results of conditional generation can be found in Appendix C.

C. UNSUPERVISED CLUSTERING

Given a new whistle observation x_{new} , one can estimate which Gaussian component $\hat{k}$ it belongs to using equations (11) and (12), as previously discussed. Fig. 10 illustrates the result of this MAP-based estimation: the left panel shows the input whistle contour, and the right panel visualizes 100 whistle contours sampled from the estimated component $\hat{k}$. In the right panel, the red line represents the input whistle contour, while the blue lines indicate the sampled contours. As shown, the contour generated from the estimated component closely resemble the input contour, demonstrating that the VAE-GMM structure inherently supports unsupervised classification for new whistle data.

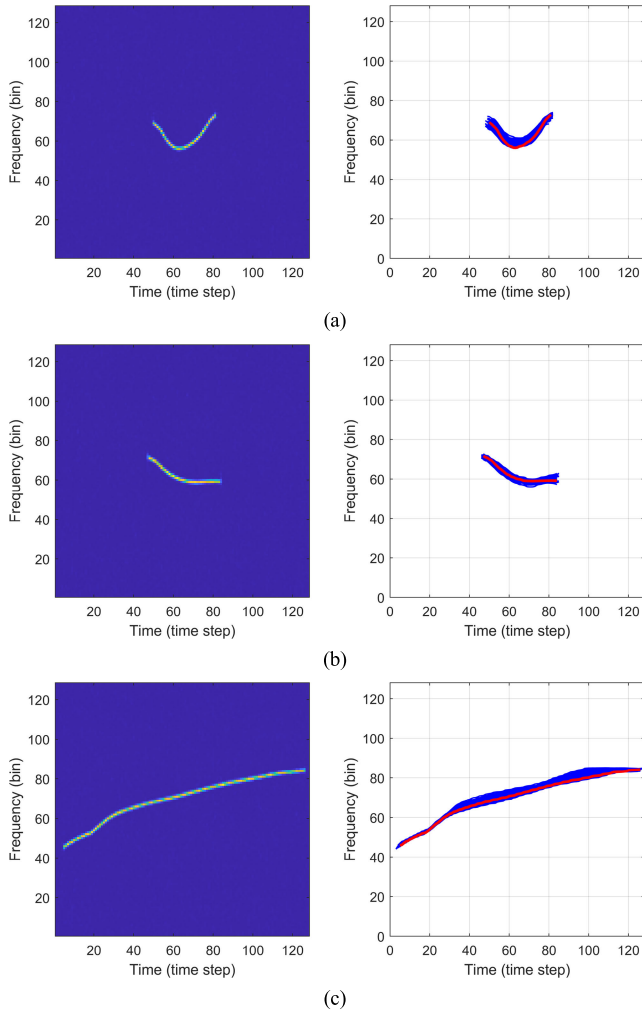


FIGURE 10. Visualization of MAP-based component estimation from new whistle input data (left panel). The right panel overlays the input contour (red line) with 100 whistle contours (blue lines) sampled from the estimated Gaussian component. (a) $k = 9, 441$, (b) $k = 5, 871$, (c) $k = 3, 087$.

TABLE 1. Performance evaluation result of VAE with WGAN-GP.

L	MSE	PSNR [dB]	SSIM	KL div.(latent)
32	0.7444	43.4275	0.9845	24.3390
64	0.5778	44.5274	0.9861	43.1929
128	0.5494	44.7466	0.9844	108.7504
256	0.3861	46.2786	0.9879	197.5745

However, component estimation via MAP inference may lead to low classification accuracy for some components. This can be attributed primarily to two reasons. First, when the Mahalanobis distance between two components in the latent space is small, misclassification may occur due to their small statistical distance. Second, in cases where the mixing coefficients are highly imbalanced, the MAP estimation may favor the component with a larger mixing coefficient, even if the Mahalanobis distance to another component is smaller.

To mitigate such issues, grouping similar components into broader categories can be an effective strategy. For example, suppose two components frequently result in

TABLE 2. Performance evaluation result based on the hyperparameters of the proposed VAE-GMM.

L	β	K	FID ₁	FID ₂	Recall [%]	Precision [%]	F1-score [%]
64	10^{-4}	1k	4.37	72.81	86.52	89.58	88.02
		5k	3.04	69.99	86.92	91.33	89.07
		10k	2.68	68.74	87.21	91.99	89.54
		15k	2.64	68.45	87.54	92.34	89.88
	10^{-2}	10k	7.20	32.62	83.29	92.25	87.54
	10^{-6}	10k	2.68	68.74	87.21	91.99	89.54
			7.98	44.00	82.17	87.14	84.58

misclassification due to the aforementioned reasons. If the whistle contours generated from these two components are visually similar, grouping them together may reduce the overall classification error. This component grouping process can be viewed as an analytical step toward a deeper understanding of the latent space structure of whistle signals.

To mitigate such issues, grouping similar components into broader categories can be an effective strategy. For example, suppose two components frequently result in misclassification due to the aforementioned reasons. If the whistle contours generated from these two components are visually similar, grouping them together may reduce the overall classification error. This component grouping process can be viewed as an analytical step toward a deeper understanding of the latent space structure of whistle signals.

D. QUANTITATIVE PERFORMANCE ANALYSIS

1) VARIATIONAL AUTOENCODER

We first evaluated the performance of the VAE across different latent vector dimensions L using four quantitative metrics: Mean Square Error (MSE), Peak Signal-to-Noise Ratio (PSNR), Structural Similarity Index Measure (SSIM), and KL divergence of the latent variables. Table 1 summarizes the evaluation results. As the latent dimensionality increased, MSE consistently decreased while PSNR showed improvements, indicating better reconstruction quality. In contrast, SSIM generally increased with the latent dimension, except for the case of $L = 128$. The decrease in SSIM at $L = 128$ appears to be due to the encoder failing to effectively capture features, resulting only in latent dispersion without the evaluation results. As the latent dimensionality increased, MSE consistently decreased while PSNR showed improvements, indicating better reconstruction quality. In contrast, SSIM generally increased with the latent dimension, except for the case of $L = 128$. The decrease in SSIM at $L = 128$ appears to be due to the encoder failing to effectively capture features, resulting only in latent dispersion without meaningful representation. KL divergence values were relatively large because the weighting coefficient β for the KL term in the overall VAE loss was set to a small value of 10^{-4} . This is in contrast to standard VAEs where $\beta = 1$.

Furthermore, the KL divergence increased with latent dimensionality, which can be interpreted as the latent variables being more widely distributed when the dimensionality increases while β remains constant.

Based on the trends in MSE and KL divergence shown in Table 1, we primarily used the VAE model with latent dimension 64 for visualization of whistle generation results and performance evaluation in this study.

2) VAE-GMM WHISTLE GENERATIVE MODEL

To quantitatively assess the generation quality of the proposed VAE-GMM and its similarity to the real whistle dataset, we adopted the FID, Recall, Precision, and F1-score. FID is a widely used statistical metric for evaluating the fidelity and diversity of synthesized images [36]. It computes the distance between real and generated image distributions by comparing the 2,048-dimensional activation vectors from the final pooling layer of a pre-trained Inception V3 network. Lower FID values indicate higher similarity between the real and generated data. Recall, Precision, and F1-score are quantified following the method proposed in [47].

We computed FID using two approaches. The first method, denoted as FID_1 , binarized both the real and generated images using a threshold of 0.1 before inputting them into the Inception V3 network. This approach aims to mitigate performance distortion caused by background noise or irrelevant visual details unrelated to the generation of whistle contour. The second method, FID_2 , fed the original grayscale images directly into the network without any thresholding, which is the more standard practice in FID evaluation. However, because the shape and diversity of whistle contours are the primary focus in our study, we considered FID_1 to be a more meaningful evaluation metric.

Since the Inception V3 model expects RGB input images of size (224,224,3), we resized the original 128×128 grayscale images using bicubic interpolation and replicated the single channel to create a three-channel RGB input.

Table 2 reports the FID scores, Recall, Precision, F1-score for the VAE-GMM when the number of GMM components was set to 1,000, 5,000, 10,000 and 15,000. The results show that increasing the number of components consistently improved the metrics. All performance metrics showed the most improved results when $K = 15,000$.

Table 2 also summarizes the performance analysis results based on the β weight of the KL divergence term in the loss function. Focusing on FID_1 and F1-score, the performance was overall superior when $\beta = 10^{-4}$ compared to when $\beta = 10^{-2}$, and $\beta = 10^{-6}$.

It is also notable that there was a substantial difference between FID_1 and FID_2 , which implies that fine image details significantly influence FID. Given that the main objective of this work is to generate accurate time-frequency contours rather than detailed pixel-level images, we argue that FID_1 —excluding background and irrelevant variations—is a more appropriate performance indicator for our application.

3) UNSUPERVISED CLUSTERING

In this section, we evaluate the quality of unsupervised clustering using three performance metrics: the Silhouette score, Davies-Bouldin Index (DBI), and Calinski-Harabasz (CH)

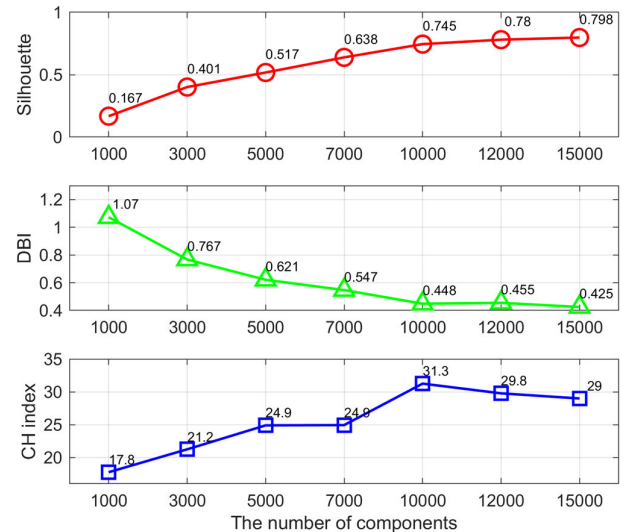


FIGURE 11. Variations in Silhouette score, Davies-Bouldin Index (DBI), and Calinski-Harabasz (CH) index relative to the number of GMM components. The Silhouette score exhibited a monotonic increase with K , while the DBI converged at $K=10,000$. Conversely, the CH index reached its peak at $K=10,000$ and declined thereafter. A comprehensive evaluation of these three metrics confirms that a component count between 10,000 and 12,000 is optimal for the model.

index [44], [45], [46]. These metrics collectively measure both intra-cluster compactness and inter-cluster separation. In particular, we analyze the performance of unsupervised clustering with respect to the number of GMM components to establish a criterion for selecting an appropriate number of clusters.

Although all three metrics assess clustering quality, their calculation methods differ. The Silhouette score measures the normalized difference between the average distance within the assigned cluster and that to the nearest neighboring cluster; thus, a higher value indicates better clustering. In contrast, DBI is computed as the ratio of cluster compactness to the distance between cluster centers, where a lower value is preferred. The CH index is defined as the ratio of between-cluster variance to within-cluster variance, with higher values indicating better performance.

Fig. 11 shows the variations of the Silhouette score, DBI, and CH index according to the number of GMM components. As the number of components increases, the Silhouette score increases monotonically, but the rate of increase becomes marginal when $K \geq 12,000$. DBI reaches its minimum at $K = 15,000$, although it remains comparable to the values at $K = 10,000$ and $K = 12,000$. The CH index increases proportionally with K , peaks at $K = 10,000$, and then decreases as K continues to grow.

Based on these observations, the appropriate number of components for unsupervised clustering of whistle data can be determined to lie between $K = 10,000$ and $K = 12,000$, where the CH index achieves its maximum value.

4) CONDITIONAL GENERATION

We evaluated the conditional generation performance from two perspectives. First, we assessed the statistical

TABLE 3. Performance evaluation for conditional generation of whistle images.

No.	Generation Condition	c_1	c_2	N	1D Wasserstein-1 (Average error)	Fréchet Feature Distance (FFD)	Recall [%]	Precision [%]	F1-score [%]
1	$c_1 \leq L \leq c_2$ [sample]	30	40	30,593	1.0075 [ms]	0.0899	98.94	83.95	90.83
2		75	80	8,202	1.7744 [ms]	1.4866	99.33	83.19	90.54
3		88	108	21,409	1.2985 [ms]	0.7703	99.18	87.31	92.87
4	$c_1 \leq f_{min} \leq c_2$ [bin]	34	55	33,972	5.9156 [Hz]	1.1677	98.71	88.07	93.09
5		40	53	21,056	3.4688 [Hz]	1.2994	98.81	89.03	93.66
6	$c_1 \leq f_{max} \leq c_2$ [bin]	62	66	28,503	3.3562 [Hz]	0.2531	99.07	85.37	91.71
7		68	71	58,918	1.4625 [Hz]	0.0599	99.01	85.08	91.51
8	$c_1 \leq \Delta f \leq c_2$ [bin]	10	20	59,624	2.8969 [Hz]	0.1105	99.04	84.96	91.46
9		20	30	22,851	2.9156 [Hz]	0.3384	98.69	84.93	91.29
10	$c_1 \leq f_{start} \leq c_2$ [bin]	40	55	19,354	3.9938 [Hz]	0.6993	98.71	84.55	91.08
11		70	80	27,062	3.5906 [Hz]	0.2902	98.74	86.04	91.95
12	$c_1 \leq f_{end} \leq c_2$ [bin]	40	55	10,712	5.6063 [Hz]	1.1084	98.17	90.84	94.36
13		70	80	48,642	3.5156 [Hz]	0.1313	98.93	82.66	90.07

similarity between real and synthetic whistle data that satisfy a given condition, which allows us to determine how faithfully the generated samples adhere to the specified condition. Second, we examined the statistical similarity between real and synthetic data across different conditions. By comparing these results with the evaluation metrics obtained under the same condition, we can indirectly verify that the proposed VAE-GMM-based conditional whistle generation meaningfully reflects the underlying structure of the real data distribution.

Performance Evaluation 1: We evaluated the statistical similarity between conditionally generated samples and real whistle data in the feature domain. For this purpose, we employed five quantitative metrics: 1D Wasserstein-1, Fréchet Feature Distance (FFD), Recall, Precision, and F1-score. We conducted experiments under two distinct conditions for six whistle characteristics: duration, minimum frequency, maximum frequency, bandwidth, start frequency, and end frequency. For evaluation of conditional generation, we used a VAE trained with $L = 64$, $K = 10,000$, and $\beta = 10^{-4}$.

To statistically evaluate the similarity between conditionally generated whistles and real data in the feature domain, we conducted the following analysis procedure:

- 1) **Real dataset preparation:** We prepared a real whistle dataset X that satisfies the target generation condition. This dataset contains N whistle images.
- 2) **Synthetic dataset generation:** Under the same condition, we generated a synthetic dataset Y , also consisting of N whistle images.
- 3) **Feature extraction:** For both datasets, we extracted feature vectors X_f and Y_f . Each vector consists of seven elements: duration L , minimum frequency f_{min} , maximum frequency f_{max} , bandwidth Δf , start frequency f_{start} , end frequency f_{end} , and center frequency f_c . These features were obtained by converting each whistle image into its time-frequency representation as illustrated in Fig. 2(c).
- 4) **Conditional feature similarity:** To evaluate similarity specifically for the conditioned feature, we computed

the 1D Wasserstein-1 distance between the corresponding real and generated feature distributions. This metric measures the average difference only for the conditioned feature. For example, when the condition is $30 \leq L \leq 40$, only the duration distributions of X and Y are compared, while the other features are ignored.

- 5) **Full feature-space similarity:** To examine the overall consistency in the entire 7-dimensional feature space, we computed FFD, Recall, Precision, and F1-score. Unlike Wasserstein-1, these metrics capture global statistical similarity between real and generated samples across all features, not only those used as conditions.

The one-dimensional Wasserstein-1 distance between two sample sets $X = \{x_{(i)}\}_{i=1}^N$ and $Y = \{y_{(i)}\}_{i=1}^N$ can be computed in closed form as:

$$W_1(X, Y) = \frac{1}{N} \sum_{i=1}^N |x_{(i)} - y_{(i)}|, \quad (13)$$

where $x_{(i)}$ and $y_{(i)}$ denote the i -th order statistics obtained by sorting the samples in X and Y in ascending order, respectively. In other words, the Wasserstein-1 distance in one dimension corresponds to the average absolute difference

between the aligned quantiles of the two empirical distributions. The FFD is computed in the same manner as the FID but applied to the 7-dimensional feature vectors. Recall, Precision, and F1-score are also calculated according to the procedure presented in Table 2.

Table 3 summarizes the evaluation results obtained from this procedure. Because the center frequency of all whistles is normalized to the center of the image, it was excluded from conditional inputs. Accordingly, we defined several conditions for each of the remaining six features— L , f_{min} , f_{max} , Δf , f_{start} , and f_{end} .

The results indicate that the conditionally generated whistles exhibit statistically comparable similarity to real whistle data. For duration-based conditions, the average error was within a few milliseconds. For frequency-related conditions (f_{min} , f_{max} , Δf , f_{start} , and f_{end}), the mean errors ranged from 1.46 Hz to 5.92 Hz, demonstrating close alignment with real data. The FFD values (0.0899-1.4866) also indicate good

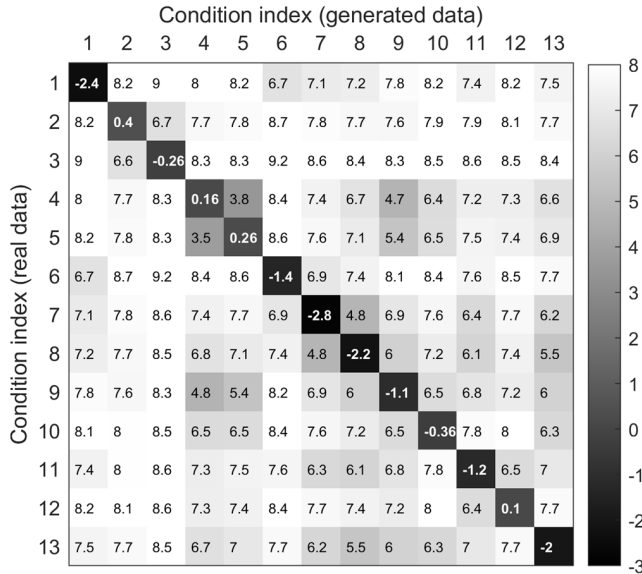


FIGURE 12. Confusion matrix of the log-transformed FFD values computed between real whistle data and conditionally generated samples produced by the VAE-GMM. The condition index corresponds to the condition numbers summarized in Table 3. This figure provides an integrated view of the statistical differences between real and synthetic whistle data across various conditions. The diagonal elements exhibit smaller values compared to the off-diagonal entries, indicating that the statistical discrepancy between real and synthetic data is relatively small when evaluated under the same condition.

TABLE 4. Performance evaluation for ablation study.

Model	FID ₁	FID ₂	Recall [%]	Precision [%]	F1-score [%]
Standard VAE (L=64)	91.83	502.41	23.70	37.51	29.05
WGAN-GP	13.82	202.51	61.47	83.17	70.70
VAE-GMM (w/o WGAN-GP, L=64, K=10k)	7.37	182.84	80.58	92.96	86.33
VAE-GMM (w/ WGAN-GP, L=64, K=10k)	2.68	68.74	87.21	91.99	89.54

similarity, and all F1-scores exceeded 90%. Recall values were particularly high, reflecting the model's ability to generate diverse whistle patterns. Consistent with the FID analysis, Precision values were slightly lower than Recall.

Performance Evaluation 2: To examine the statistical differences between real and synthetic whistle data under various conditions, we computed the FFD matrix. Fig. 12 shows the log-transformed FFD matrix, defined as $Z(i, j) = \log(\text{FFD}(i, j))$, for the 13 condition indices presented in Table 3. Here, i denotes the condition index of the real data and j denotes that of the synthetic data, corresponding to the x- and y-axes of Fig. 12, respectively. For example, $Z(4, 5)$ represents $\log(\text{FFD}(4, 5))$, which measures the statistical distance between real whistles satisfying condition 4 ($34 \leq f_{\min} \leq 55$) and synthetic whistles generated under condition 5 ($40 \leq f_{\min} \leq 53$). The detailed definitions of all condition indices can be found in Table 3, and the diagonal entries of Fig. 12 correspond exactly to the $\log(\text{FFD})$ values reported in Table 3.

As shown in Fig. 12, the diagonal entries exhibit smaller values than the off-diagonal ones. This indicates that the statistical discrepancy between real and synthetic data under the same condition is relatively small compared to that between different conditions. Such a relative comparison provides indirect evidence that the VAE-GMM generates whistle data consistent with the intended condition, demonstrating its effectiveness in conditional generation.

An interesting observation in Fig. 12 is that the FFD values for conditions 4 and 5 are noticeably smaller than other off-diagonal elements. Specifically, $Z(4, 5) = 3.8$ and $Z(5, 4) = 3.5$, whereas other off-diagonal values range from 4.7 to 9.2. This behavior arises because the minimum-frequency range of condition 5 is a subset of that of condition 4, making the two datasets inherently similar. Consequently, the relatively low values of $Z(4, 5)$ and $Z(5, 4)$ are consistent with the statistical relationship between these conditions. These results collectively demonstrate that the FFD metric sensitively captures feature-space similarity and that the conditional generation of the VAE-GMM is highly effective, as evidenced by the structure of Fig. 12.

5) ABLATION STUDY

The proposed whistle generation VAE-GMM consists of three key components: a VAE, a GMM, and a WGAN-GP. To verify the contribution of each module, we conducted an ablation study by comparing four configurations: (1) a standard VAE, (2) WGAN-GP, (3) a VAE-GMM without WGAN-GP, and (4) a VAE-GMM with WGAN-GP. This comparison allows us to evaluate the effectiveness of multimodal distribution modeling using GMM and the fine-quality enhancement effect achieved by WGAN-GP.

Table 4 summarizes the FID, Recall, and Precision results for the four network configurations. As shown in the table, the VAE-GMM with WGAN-GP achieved the best performance, with an FID of 2.6767, Recall of 87.21%, and Precision of 91.99%. Furthermore, as GMM and WGAN-GP were progressively incorporated into the standard VAE, the overall performance improved consistently. The superior performance of the VAE-GMM compared to the basic VAE indicates its effectiveness in capturing greater diversity in the latent variables. Additionally, the further improvement observed after applying WGAN-GP suggests that the critic indicates its effectiveness in capturing greater diversity in the latent variables. Additionally, the further improvement observed after applying WGAN-GP suggests that the critic network of WGAN-GP contributes to enhancing the fine-grained quality of the generated data.

6) PERFORMANCE COMPARISON

To compare the performance of the proposed method with other generative models, we considered models trainable on the false killer whale whistle dataset we constructed. As illustrated in Fig. 1, the whistle data built in this study exhibits the characteristics of sparse data, where the majority of pixel values are zero. This sparsity consequently posed difficulties in

TABLE 5. Performance comparison of existing methods and VAE-GMM.

Model	FID ₁	FID ₂	Recall [%]	Precision [%]	F1-score [%]
VAE-GMM ($L=64, K=10k$)	2.68	68.74	87.21	91.99	89.54
VAE-GMM ($L=64, K=15k$)	2.64	68.45	87.54	92.34	89.88
WhistleGAN	25.43	118.03	72.25	41.79	52.95
WGAN-GP	13.82	202.51	61.47	83.17	70.70
LDM, $T=250$	9.65	82.92	84.04	90.28	87.05
LDM, $T=500$	6.60	87.34	83.03	92.13	87.34
LDM, $T=1,000$	5.92	83.09	83.30	91.57	87.24

TABLE 6. Computational cost and generation speed.

Model	Params	FLOPs	Infer. latency	Per-image latency $B=16$	$B=32$
VAE-GMM, $L=64$	368k	557M	1.7 ms	0.11 ms	54 us
WhistleGAN	7.7M	3.79G	1.8 ms	0.11 ms	57 us
WGAN-GP	5.1M	1.71G	1.6 ms	0.10 ms	52 us
LDM, $T=250$	2.9M	23.8G	1.80 s	0.16 s	0.13 s
LDM, $T=500$	2.9M	47.5G	3.62 s	0.32 s	0.25 s
LDM, $T=1,000$	2.9M	95.0G	7.23 s	0.64 s	0.32 s

TABLE 7. Network architecture of variational autoencoder.

Module	Kernel / Stride / # of Filters	Output Size
Input	-	128×128×1
Encoder 1 (conv, swish)	3 / 2 / 32	64×64×32
Encoder 2 (conv, swish)	3 / 2 / 32	32×32×32
Encoder 3 (conv, swish)	3 / 2 / 64	16×16×64
Encoder 4 (conv, swish)	3 / 2 / 64	8×8×64
Fully connected layer	4096×2L / · / ·	2L×1
Sampling layer	-	L×1
Projection and reshape	L×4096 / · / ·	8×8×64
Decoder 1 (transposed, swish)	3 / 2 / 64	16×16×64
Decoder 2 (transposed, swish)	3 / 2 / 64	32×32×64
Decoder 3 (transposed, swish)	3 / 2 / 32	64×64×32
Decoder 4 (transposed, swish)	3 / 2 / 32	128×128×32
Output (sigmoid)	3 / 1 / 1	128×128×1

TABLE 8. Network architecture of critic network.

Module	Kernel / Stride / # of Filters	Output Size
Input	-	128×128×1
Conv, leakyRelu(0.2)	5 / 2 / 32	64×64×32
Conv, layerNorm, leakyRelu(0.2)	5 / 2 / 32	32×32×32
Conv, layerNorm, leakyRelu(0.2)	5 / 2 / 64	16×16×64
Conv, layerNorm, leakyRelu(0.2)	5 / 2 / 64	8×8×64
Output (sigmoid)	8 / 1 / 1	1

training, depending on the deep learning model architecture used.

Conventional adversarial models, such as WhistleGAN (DCGAN) [20], [48] and WGAN-GP [49], were unable to independently generate valid whistle data. These models consistently yielded suboptimal results, primarily outputting near zero-pixel values that lacked meaningful structures. To mitigate this vanishing gradient issue, we re-trained the WhistleGAN from scratch by incorporating Label

TABLE 9. Network architecture of diffusion model.

Module	Kernel / Stride / # of Filters	Output Size
Input	-	8×8×1
ConvBlock 1	3 / 1 / 64	8×8×64
Downsample 1	3 / 2 / 64	4×4×64
ConvBlock 2	3 / 1 / 128	4×4×128
Downsample 2	3 / 2 / 128	2×2×128
ConvBlock 3	3 / 2 / 256	2×2×256
Upsample 1	3 / 2 / 256	4×4×(256+128) *Depth concatenation with convBlock 2 output
ConvBlock 4	3 / 1 / 128	4×4×128
Upsample 2	3 / 2 / 128	8×8×(128+64) *Depth concatenation with ConvBlock 1 output
ConvBlock 5	3 / 1 / 128	8×8×64
Output (conv)	1 / 1 / 1	8×8×1

*ConvBlock: Embedding addition, convolution, group norm, swish layer

*Downsample: Convolution layer

*Upsample: Transposed convolution, depth concatenation layer

Smoothing and Label Flipping techniques. These adjustments were specifically designed to prevent the premature saturation of the Discriminator, thereby facilitating the emergence of valid whistle contours.

In contrast, the Latent Diffusion Model (LDM)—which generates the latent representations of a VAE through a diffusion process—demonstrated the capability to generate high-fidelity whistle sounds without any additional heuristic manipulations. Detailed information regarding the neural network architecture and training methodology of the LDM is provided in Appendix D.

Table 5 presents a comparative performance analysis of VAE-GMM, WhistleGAN, WGAN-GP, and LDM. The comparison indicates that the VAE-GMM achieved superior performance compared to other models. In Table 5, T denotes the number of time-steps in the LDM, where performance generally improves as T increases. The superior performance of VAE-GMM over LDM is likely due to the fact that the Diffusion Model's U-Net backbone is typically optimized for effective noise removal in high-dimensional tensors, which are usually larger than 64 dimensions.

7) COMPUTATIONAL COST

We evaluated the computational cost of each whistle sound generation model based on the number of learnable parameters, Floating-Point Operations Per Second (FLOPs), single-image inference latency, and batch latency. For the computational cost assessment, we only considered the neural network components that directly contribute to image generation. For instance, modules such as the WGAN-GP discriminator or the VAE encoder, which are not essential for the final generative process, were excluded from the cost calculation.

In calculating the FLOPs for each model, we solely considered the layer types that constitute a significant portion of the total operations. The layers included in the FLOPs computation were the fully connected layer, transposed-convolution,

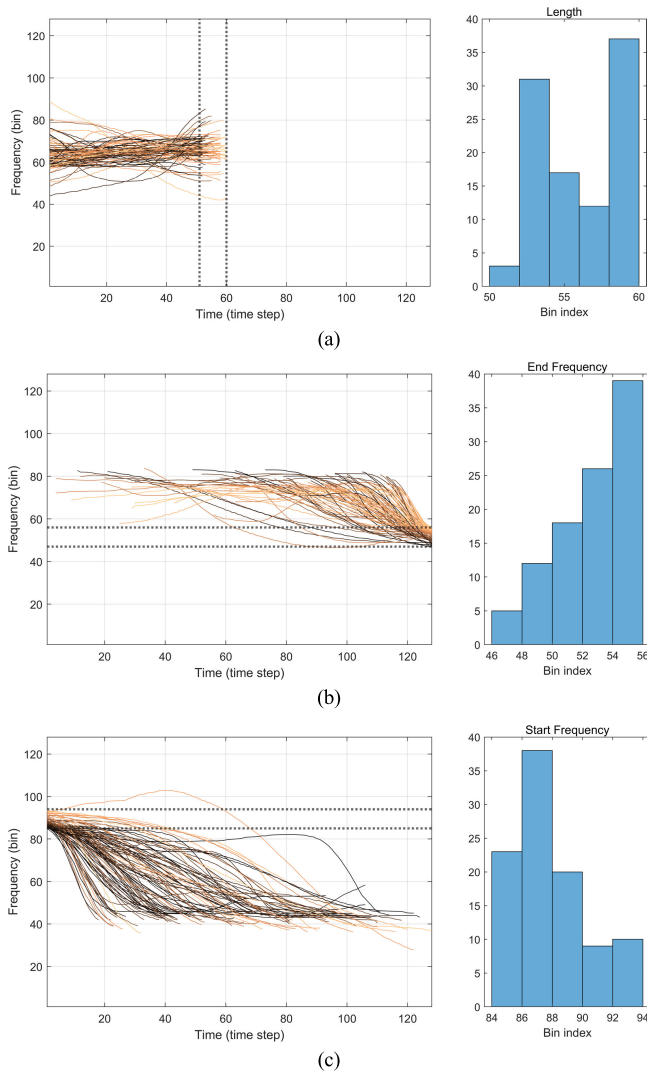


FIGURE 13. Conditional whistle generation results 1. (a) Length condition: 51-60 samples, (b) End frequency condition: 47-56 bin index, (c) Start frequency condition: 85-94 bin index.

batch normalization, ReLU, sigmoid, and max-pooling layer; all other layers were intentionally omitted.

To ensure a fair comparison, the generation speeds of all models listed in Table 6 were evaluated under identical hardware environments and optimization settings. The evaluation was performed on a desktop system equipped with an Intel Core i7-13700 CPU, an NVIDIA GeForce GT 1030 GPU, and 64 GB of RAM. The inference and batch latencies were calculated by averaging 200 trials, excluding the initial 50 warm-up runs out of a total of 250 attempts. To ensure measurement stability and consistent thermal states, a 0.5-second settling time was introduced between consecutive generation cycles. All primary computations were configured to be executed via GPU acceleration.

As presented in Table 6, the VAE-GMM achieved an inference latency of 1.7 ms for a single image, and remarkable throughputs of 0.11 ms and 54 μ s for batch sizes of 16 and 32, respectively. The computational speed of the VAE-GMM was found to be comparable to those of WGAN-GP and

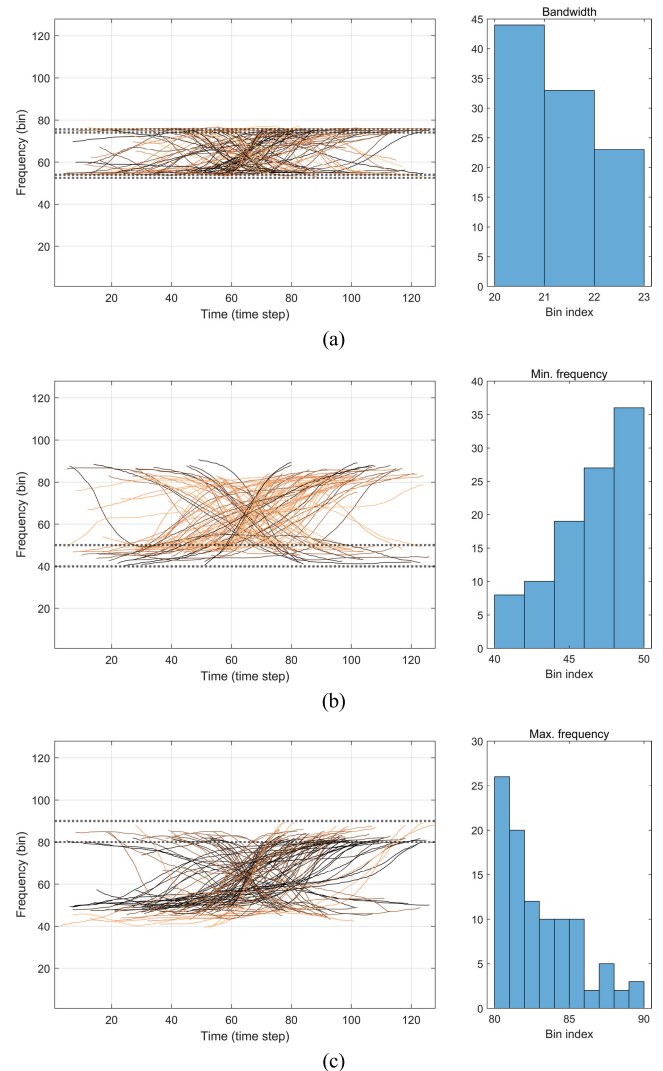


FIGURE 14. Conditional whistle generation results 2. (a) Bandwidth condition: 20-23 bins, (b) Minimum frequency condition: 40-50 bin index, (c) Maximum frequency condition: 80-90 bin index.

WhistleGAN. This comparable performance is attributed to the architectural similarity between the VAE decoder and the GAN generators used in this study.

The LDM incurs a significantly higher computational overhead than other models because it generates data by iteratively refining the noisy latent variable via a U-Net. In terms of FLOPs, LDM necessitates a minimum of 42.63 times to a maximum of 170.49 times more operations than the VAE-GMM.

Although the computational cost of LDM can be alleviated by reducing the total number of time-steps T , it still maintains a substantially higher computational load than VAE-GMM. Given that the FLOPs per time-step of the U-Net is 95.02 MFLOPs, even a reduction of the time-steps to $T = 100$ results in LDM's total FLOPs being 17.05 times greater than those of VAE-GMM. However, decreasing the number of time-steps in LDM inherently sacrifices generation quality, rendering this approach unsuitable. As confirmed in Table 5,

the image generation quality of LDM is highly sensitive to the number of time-steps.

V. DISCUSSIONS

The proposed VAE-GMM-based whistle generation method has two major limitations, as described below:

First, whistle signal generation currently requires a four-step conversion process—from whistle image generation to waveform synthesis. Since our ultimate goal is to generate whistle waveforms that can be directly transmitted through an underwater acoustic projector, a more efficient future approach would be to train a model capable of generating the transmit waveform directly.

Second, the proposed VAE-GMM learns only the contour shape of whistles with their center frequencies removed. To transmit signals through an actual underwater acoustic projector, an additional method for assigning center frequencies is required. We attempted to train the model using spectrogram images that include the center frequency; however, this approach was found to be ineffective.

Specifically, when the center frequency was not removed—meaning the whistle contour could shift vertically within the image—the data variability increased substantially. As a result, the VAE reconstruction loss was sufficiently reduced only when the latent dimension was set to much higher values (e.g., 128 or 256) compared to the proposed 64-dimensional configuration. However, when the latent dimension exceeded 128, we found that modeling the latent space with much more than 10,000 Gaussian components became computationally impractical due to the computational complexity of the EM algorithm.

Therefore, we expect that assigning the center frequency of the data contributing to each Gaussian component and sampling the center frequency from this distribution would serve as an effective alternative. We leave a detailed investigation of this issue for future work.

VI. CONCLUSION

This paper proposed a VAE-GMM-based method for generating whistle image data of false killer whale. The proposed approach explicitly models the latent space of whistle images as a mixture of Gaussians, enabling conditional generation through analysis of component-wise features, such as minimum and maximum frequency, start and end frequency, bandwidth, center frequency, and duration. It was shown that the conditional probability distribution can be approximately estimated based on the characteristics of the selected Gaussian components. Furthermore, the analytical tractability of the mixture density enables the use of MAP estimation for potential applications in unsupervised clustering and classification.

The proposed VAE-GMM supports fast whistle data generation, comparable to GAN-based methods, while offering an intuitive mechanism to control the generation process through the mixture components. The method demonstrated competitive performance, suggesting its effectiveness for future

applications in biomimetic underwater communication systems.

We anticipate that the proposed VAE-GMM-based whistle generation method will serve as an effective tool for the linguistic analysis and understanding of cetacean acoustic signals. A systematic method for structuring these signals is essential for their linguistic interpretation. While conventional studies have predominantly relied on manual pattern analysis through direct observation, our approach offers a novel framework for structuring whistle morphologies in an unsupervised manner. We believe that this method will provide a foundational basis for future research, such as identifying correlations between distinct whistles and establishing links between acoustic signals and their behavioral contexts.

APPENDIX A LEARNING VARIATIONAL AUTOENCODER

A. NETWORK ARCHITECTURE

The network architectures of the VAE and the critic, depending on the latent space dimension $\mathcal{L}$, are summarized in Tables 7 and 8, respectively.

B. TRAINING CONFIGURATION AND NETWORK ARCHITECTURE

The training of VAE-GMM was conducted using MATLAB R2024a, and computations were performed in parallel across seven NVIDIA V100 GPUs.

For VAE training, the learning rate was set to 10^{-3} , and the β coefficient was set to 10^{-4} . The mini-batch size was 128 per GPU, resulting in a total of 896 samples per batch. The WGAN-GP weight λ was set to 1, and the critic was trained three times per iteration of the VAE. The total number of training epochs was 300. The gradient penalty coefficient λ_{GP} was set to 5. The learning rate for the critic was 2×10^{-4} , with a gradient decay factor of 0 and a squared gradient decay factor of 0.9. The Adam optimizer was employed for training both the VAE and the critic network.

During the training process, a data augmentation technique was applied in which each image was randomly rotated by 0° , 90° , 180° , or 270° with equal probability. This approach was adopted to prevent the VAE from overly focusing on horizontal line features, which can be prevalent in whistle image data, and to encourage balanced learning of diverse directional features, such as vertical lines.

APPENDIX B PARAMETER ESTIMATION OF GAUSSIAN MIXTURE MODEL

The parameters of the Gaussian mixture model consist of the mean vectors, covariance matrices, and mixing coefficients corresponding to the specified Gaussian components. These parameters are estimated using the EM algorithm. In this study, the maximum number of EM iterations was set to 1,500, and the probability tolerance for early stopping was set to 10^{-8} .

APPENDIX C

CONDITIONAL WHISTLE GENERATION RESULTS

Fig. 13 and 14 show the results of conditional whistle generation under six different conditions—whistle length, end/start frequency, maximum/minimum frequency, and bandwidth. For each case, 100 generated whistle samples are overlaid in the left panel, and the corresponding histogram of the target feature is shown in the right panel. The center frequency was excluded from the conditional generation results, as it was normalized during the preprocessing of the whistle dataset used in this study.

APPENDIX D

IMPLEMENTATION OF THE LATENT DIFFUSION MODEL

To compare the performance of the proposed VAE-GMM-based whistle generation method with a widely recognized generative framework, we applied the LDM to the whistle image dataset constructed in this study. The U-Net backbone of the LDM follows the network architecture presented in Table 9 and takes an $8 \times 8 \times 1$ latent image as input to predict the noise residual. The latent vectors used for diffusion-based generation were extracted from a VAE trained with a KL divergence weight of 10^{-4} .

For training the diffusion model, the hyperparameters were configured as follows: a minibatch size of 128, 300 training epochs, and a learning rate of 0.0005. The variance schedule parameters were set to $\beta_{min} = 10^{-4}$ and $\beta_{max} = 0.02$, with a linear schedule applied. The U-Net was trained on a desktop equipped with an NVIDIA GeForce GT 1030 GPU, an Intel Core i7-13700 CPU, and 64 GB of RAM.

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